

Test Report - VxScan Thermals with Updated Hardware/Software

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Summary

This investigation verified improved temperatures in VxScan v4.0.4 as of March 2026 under higher stress operating conditions. The tested unit included hardware that circulated heat away from the internal SBC and software that reduced processing by the SBC and SSD. All measured temperatures were within specifications for internal components and within safe touch limits for external surfaces.

Applicable VVSG Requirements

- 8.1-K – Eliminating hazards

Devices Under Test

- VxScan, v4.0, Build 0.5.2, image v4.0.4-rc2, all-bubble ballot
- SanDisk 64GB Ultra Flair USB3.0 USB drive

Purpose

This investigation intended to verify improvements in heat management in VxScan. Previous thermal testing on an early VxScan prototype suggested the need to reduce heat accumulating around the single-board computer (SBC) and solid-state drive (SSD) on the left side of the unit, as reported in *Test Report - VxScan Thermals*. Updates were made to both hardware and software to address this. In hardware, a heatsink with a fan was integrated with the SBC that circulated air away from that compartment better. In software, code changes reduced memory usage by the SBC/SSD and reduced reported chip temperatures, as documented in [VxSuite GitHub issue #6699](#). New measurements while running under higher stress conditions would be compared to the previous measurements.

Two different SSD products were also compared for thermal behavior: an ADATA 128GB model (70°C rated maximum operating temperature) and a Kingston 256GB model (85°C rated maximum operating temperature). The ADATA SSD was used in the original prototype, and the

Kingston model was considered for a newer updated design. Measurements were taken to verify acceptable thermal management with either SSD.

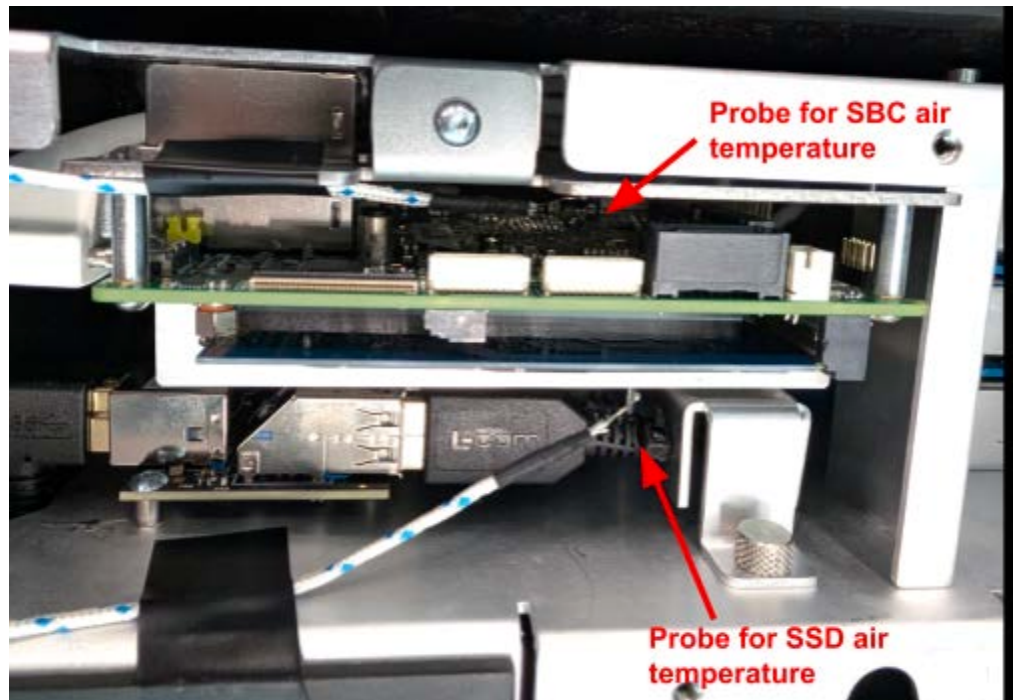
Materials

- VxScan unit
 - Imaged with v4.0.4-rc2 production app
 - Configured with all-bubble ballot election, continuous scanning “shoeshine” mode
 - Mounted on ballot box
- VxScan peripherals
 - Test paper, all-bubble ballot
 - USB drive
 - Smart card
 - Thermal paper roll
- Tools for accessing VxScan internals
- Thermal camera: Topdon TC004
- Thermocouple system: 4-channel system, with K-type thermocouples
- Electrical tape, scissors
- SSD samples
 - ADATA IM2P32A8-128GCTB5
 - Kingston OM8SGP4256K2-A00
- Ambient thermometer
- Climate tent
- Space heater

Procedures

1. Define key locations to measure temperatures. Internal air temperatures can be measured via thermocouple. Surface temperatures can be measured via thermocouple or thermal camera. For this test, the key locations were:
 - a. Internal air: Between SBC board and left bracket, ~1" deep from panel
 - b. Internal air: Between SSD and right bracket, ~1" deep from panel
 - c. External surface: Panel above SBC
 - d. External surface: USB port with USB3.0 stick, bracket
 - e. External surface: USB stick in port
 - f. Internal surface: SBC header
 - g. Internal surface: SSD chip
 - h. Internal surface: SSD board
2. Measure thermals of VxScan with the ADATA SSD, in typical and warm conditions.
 - a. Attach the VxScan unit to the ballot box at room temperature. Record the ambient temperature.

- b. Remove the fasteners from the bottom tub panels.
- c. Ensure the ADATA SSD is properly inserted and secured to the SBC.
- d. Affix thermocouples to the left compartment to measure the air temperature near the SBC and SSD as shown, without the probes touching the boards or brackets:



- e. Place the lower tub panel back on the VxScan over the thermocouples without fastening them back down, and monitor the ambient temperature. The setup is shown here:



- f. Plug in and power on the unit.
- g. Measure temperatures at key locations after the system is on and idle for at least 30 minutes.

- h. Configure the election for continuous operation in “shoeshine” mode. Insert the USB drive, smart card, printer paper, and ballot paper, and start continuous operation.
 - i. Measure temperatures at key locations after the system is operational (scanning) for at least 5 minutes, and any other conditions of interest.
 - j. Set up the ambient temperature to be 95°F (35°C) around the test unit as per VVSG2.0 rated maximum operational temperatures. *Note: Follow safety recommendations in setting up and using heating sources and climate tents.*
 - k. Measure temperatures at key locations as in the steps above, after sitting idle and when operational.
 - l. When measurements are complete, turn off and disconnect the system and remove all peripherals and test equipment.
3. Measure thermals of VxScan with the Kingston SSD, in typical and warm conditions.
 - a. Repeat Step 2 above but replacing the ADATA SSD with the Kingston SSD.
 - b. Additional measurements were taken after leaving the unit on for multiple days.

Results

Summary Data

Thermal measurements of VxScan with both the ADATA and Kingston and SSD models were within component specification and safe touch limits. The unit was cooler during operation than in the older VxScan prototype with older hardware and older software. Air temperatures around the SBC and SSD were approximately 27-42°F cooler (15-23°C cooler) than previously measured.

See [Appendix A](#) for measurements on the VxScan unit with the ADATA SSD, comparing data from both the newer and older versions of the hardware and software.

See [Appendix B](#) for measurements on the VxScan unit with the Kingston SSD. This also includes data when leaving the unit plugged in and on for several days with intermittent scanning.

ADATA SSD

Worst-case internal operating air temperatures were found around the SBC after more than 30 minutes of continuous operation (“shoeshine” mode) in a warm ambient 95°F environment. This temperature was 128.6°F (53.67°C), which was below the rated 140°F (60°C) maximum for the SBC.

This is also still 27.5°F cooler (15.3°C cooler) than the corresponding location in the prototype VxScan in room temperature conditions.

Worst-case user-facing surface temperatures were found at the USB stick plugged into the USB port after more than 30 minutes of continuous operation (“shoeshine” mode) in a warm ambient 95°F environment. This temperature was 130.2°F (54.56°C), which is below the 140°F limit for safe touch recommended by OSHA and ASTM C1055.

Internal surface temperatures of the SBC and SSD were measured to be less than 158°F (70°C), within safe zones for electronics before performance throttling occurs.

Kingston SSD

Worst-case internal operating air temperatures were found around the SBC after more than 30 minutes of continuous operation (“shoeshine” mode) in a warm ambient 95°F environment. This temperature was 133.4°F (56.33°C), which was below the rated 140°F (60°C) maximum for the SBC.

Worst-case user-facing surface temperatures were found at the USB stick plugged into the USB port after more than 30 minutes of continuous operation (“shoeshine” mode) in a warm ambient 95°F environment. This temperature was 135.1°F (57.28°C), which is below the 140°F limit for safe touch recommended by OSHA and ASTM C1055.

Conclusions

- The latest hardware and software for VxScan has reduced temperatures compared to earlier prototypes.
- Internal operating air temperatures are within rated limits for components. Internal electronic components run cool enough not to expect any performance throttling.
- External user-facing temperatures are within safe touch limits.
- Both ADATA and Kingston SSD components lead to acceptable thermal performance in VxScan, even in the unlikely high-stress environments of continuous operation in 95°F ambient temperatures.

Appendix A: ADATA SSD thermals

Appendix A.1: New Hardware/Software with ADATA

Thermals with ADATA SSD, updated hardware and software March 2026				Temperature measurements (degrees F)							
				Room temperature ambient			Warm ambient				
Location	User-facing?	Type	Measurement tool	Idle 30+ min	Scanning 5 min	Scanning 25+ min	Idle 30+ min	Scanning 5 min	Scanning 25+ min		Spec: Max operating (F)
Between SBC board and left bracket, ~1" down	no	air, panel closed	thermocouple	102.4	108	116.5	113.3	118.5	128.6		140 (spec)
Between SSD and right bracket, ~1" down	no	air, panel closed	thermocouple	91.8	95.4	103.3	104.9	102.8	116.9		158 (spec)
panel above SBC	yes	surface, panel closed	thermal imager	75.1	76.4	86.3	95.3	96.4	99.1		140 (OSHA)
USB port with USB3.0 stick, bracket	yes	surface, panel closed	thermal imager	72.1	76.7	77.6	88	84	86.6		140 (OSHA)
USB stick in port	yes	surface, panel closed	thermal imager	104.2	109.8	115.8	117.6	121.9	130.2		140 (OSHA)
SBC header	no	surface	thermal imager	121.2	127.6	130.4	130.5	137.8	148.2		n/a
SSD chip	no	surface	thermal imager	124.9	134.6	133.2	139	144.2	151.1		n/a
SSD board	no	surface	thermal imager	109	118.1	119.4	125.7	129	135		n/a
			date	3/6/2026	3/6/2026	3/6/2026	3/9/2026	3/9/2026	3/9/2026		
			time start	11:30 AM	12:10:30 PM	12:10:30 PM	9:10:00 AM	10:04:30	10:04:30		
			time end	12:10 PM	12:15:30 PM	12:39:54 PM	10:00 AM	10:10:13	10:37:31		
			elapsed (min)	40.00	5.00	29.40	50.00	5.72	33.02		
			scans start	n/a	0	0	0	0	0		
			scans end	n/a	not taken	506	0	98	521		
			scans/min	n/a	not taken	17.21	not taken	17.14	15.78		
			scans/sec	n/a	not taken	0.29	not taken	0.29	0.26		
			s/scan	n/a	not taken	3.49	not taken	3.50	3.80		

Appendix A.2: Old Hardware/Software with ADATA

This is adapted from previously reported temperatures in *Test Report - VxScan Thermals..*

Thermals with ADATA SSD, old hardware and software from July 2024				Temperature measurements (degrees F)							
				Room temperature ambient			Warm ambient				
Location	User-facing?	Type	Measurement tool	Idle 30+ min	Scanning 5 min	Scanning 25+ min	Idle 30+ min	Scanning 5 min	Scanning 25+ min		Spec: Max operating (F)
Between SBC board and left bracket, ~1" down	no	air, panel closed	thermocouple	144	not taken	144	not taken				140 (spec)
Between SSD and right bracket, ~1" down	no	air, panel closed	thermocouple	n/a		n/a					158 (spec)
panel above SBC	yes	surface, panel closed	thermal imager	106		113					140 (OSHA)
USB port with USB3.0 stick, bracket	yes	surface, panel closed	thermal imager	not taken		not taken					140 (OSHA)
USB stick in port	yes	surface, panel closed	thermal imager	135		145					140 (OSHA)
SBC header	no	surface	thermal imager	199		199					n/a
SSD chip	no	surface	thermal imager	199		199					n/a
SSD board	no	surface	thermal imager	not taken		not taken					n/a
			date	7/2/24-7/15/24							
			time start	not taken							
			time end								
			elapsed (min)								
			scans start								
			scans end								
			scans/min								
			scans/sec								
			s/scan								

Appendix A.3: Comparison of New vs. Old Hardware/Software with ADATA

This compares data from Appendix A.1 minus A.2.

Difference in thermals with ADATA SSD, new hardware/software minus old hardware/software				Difference in temperature measurements (degrees F)					
				Room temperature ambient			Warm ambient		
Location	User-facing?	Type	Measurement tool	Idle 30+ min	Scanning 5 min	Scanning 25+ min	Idle 30+ min	Scanning 5 min	Scanning 25+ min
Between SBC board and left bracket, ~1" down	no	air, panel closed	thermocouple	-41.6	not taken	-27.5	not taken		
Between SSD and right bracket, ~1" down	no	air, panel closed	thermocouple	not measured		not measured			
panel above SBC	yes	surface, panel closed	thermal imager	-30.9		-26.7			
USB port with USB3.0 stick, bracket	yes	surface, panel closed	thermal imager	not taken		not taken			
USB stick in port	yes	surface, panel closed	thermal imager	-30.8		-29.2			
SBC header	no	surface	thermal imager	-77.8		-68.6			
SSD chip	no	surface	thermal imager	-74.1		-65.8			
SSD board	no	surface	thermal imager	not taken		not taken			

Appendix B: Kingston SSD thermals

Appendix B.1: New Hardware/Software with Kingston

Thermals with Kingston SSD, updated hardware and software March 2026				Temperature measurements (degrees F)									
				Room temperature ambient					Warm ambient				
Location	User-facing?	Type	Measurement tool	Idle 30+ min	Idle 90+ min	Scanning 5 min	Scanning 25+ min	Idle after use 75+ min	Idle 30+ min	Scanning 5 min	Scanning 25+ min		Spec: Max operating (F)
Between SBC board and left bracket, ~1" down	no	air, panel closed	thermocouple	98	100.6	107.1	115.4	106.4	120	125.1	133.4		140 (spec)
Between SSD and right bracket, ~1" down	no	air, panel closed	thermocouple	93.5	99	105.3	111.8	106.1	109.1	107.1	116.4		185 (spec)
panel above SBC	yes	surface, panel closed	thermal imager	82.1	76.7	76.6	78.2	86.5	97.8	96.8	104.5		140 (OSHA)
USB port with USB3.0 stick, bracket	yes	surface, panel closed	thermal imager	78.9	73.7	73.1	75.2	81.5	82.4	85.7	95.8		140 (OSHA)
USB stick in port	yes	surface, panel closed	thermal imager	108	100.7	108.9	116.2	116.6	123.4	126.8	135.1		140 (OSHA)
SBC header	no	surface	thermal imager	118.8	116.5	123.8	131.9	126.1	136.7	141.9	150.1		n/a
SSD chip	no	surface	thermal imager	122.9	126.7	133.7	133.8	131.3	142.9	145.4	153.1		n/a
SSD board	no	surface	thermal imager	118.3	110.1	120	123.5	115.1	122.8	130.6	134.8		n/a
			date	3/9/2026	3/9/2026	3/9/2026	3/9/2026	3/9/2026	3/9/2026	3/9/2026	3/9/2026		
			time start	11:53 AM	11:53 AM	1:18:45 PM	1:18:45 PM	1:53:55 PM	3:27:30 AM	4:01:20 PM	4:01:20 PM		
			time end	12:35 PM	1:15:50 PM	13:23:50	1:53:55 PM	3:12:10 PM	3:59:22 PM	4:07:50 PM	4:34:43 PM		
			elapsed (min)	42.05	82.33	5.08	35.17	78.25	751.87	6.50	33.38		
			scans start	n/a	n/a	0	0	n/a	n/a	0	0		
			scans end	n/a	n/a	79	534	n/a	n/a	101	518		
			scans/min	n/a	n/a	15.54	15.18	n/a	n/a	15.54	15.52		
			scans/sec	n/a	n/a	0.26	0.25	n/a	n/a	0.26	0.26		
			sc/scan	n/a	n/a	3.86	3.95	n/a	n/a	3.86	3.87		

Appendix B.2: Extended Operation with Kingston

Thermals with Kingston SSD, updated hardware and software March 2026; with unit on for multiple days and intermittent scanning				Temperature measurements (degrees F)						
				Room temperature ambient						
Location	User-facing?	Type	Measurement tool	Idle 12+ hrs	On 1+ days	On 2+ days	On 3+ days	Idle after use 75+ min		Spec: Max operating (F)
Between SBC board and left bracket, ~1" down	no	air, panel closed	thermocouple	105.8	105.7	106.3	103.1	106.4		140 (spec)
Between SSD and right bracket, ~1" down	no	air, panel closed	thermocouple	82.1	92.3	93.5	89.6	106.1		185 (spec)
panel above SBC	yes	surface, panel closed	thermal imager	85.7	80.4	85.3	83.1	86.5		140 (OSHA)
USB port with USB3.0 stick, bracket	yes	surface, panel closed	thermal imager	73.8	74.1	78.7	77.2	81.5		140 (OSHA)
USB stick in port	yes	surface, panel closed	thermal imager	107.7	105.8	114.1	110.4	116.6		140 (OSHA)
SBC header	no	surface	thermal imager	123.4	117.8	123.4	121	126.1		n/a
SSD chip	no	surface	thermal imager	127.9	127.4	128.9	126.6	131.3		n/a
SSD board	no	surface	thermal imager	111.2	111.2	112.1	109.9	115.1		n/a
			date	3/9-3/10/26	3/11/2026	3/12/2026	3/13/2026	3/9/2026		
			time start	3/9/2026 5:09:00 PM	3/9/2026 5:09:00 PM	3/9/2026 5:09:00 PM	3/9/2026 5:09:00 PM	1:53:55 PM		
			time end	3/10/2026 10:29:00 AM	3/11/2026 8:43:00 AM	3/12/2026 10:26:20 AM	3/13/2026 9:34:20 AM	3:12:10 PM		
			elapsed (hrs)	17.33	39.57	65.29	88.42	78.25		
			scans start	0	0	0	0	n/a		
			scans end	0	238	903	1142	n/a		